Prodigy Anesthesia Quick Review

Extracranial Surgery

Why would a parotidectomy be performed?	A parotidectomy may be performed for infection of the gland or for parotid tumor removal. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 224.*
What is the difference between a total parotidectomy and a radical parotidectomy?	A total parotidectomy spares the facial nerve, while a radical parotidectomy removes the facial nerve along with the parotid gland. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 224-225.*
What is the etiology of parotid tumors?	75% of the tumors are benign pleomorphic adenomas. The remainder are either malignant tumors or result from chronic sialoadenitis. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 226.*
How is the patient positioned for a parotidectomy?	The patient is positioned supine with the head turned slightly towards the opposite side. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 225.*
How should the endotracheal tube be positioned for a parotidectomy?	The endotracheal tube should be taped to the opposite side of the mouth from the surgical site. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 225.*
Is surgical paralysis necessary for patients undergoing parotidectomy?	No. Because the facial nerve has branches that invest the parotid gland and nerve monitoring is employed to protect the nerve, muscle relaxation is contraindicated in this procedure. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 225.*
For what reasons are enucleations performed?	An enucleation is the removal of the entire eye and a portion of the optic nerve. It is performed for intraocular tumors or painful blind eyes. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 168.*

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Extracranial Surgery

Many clinicians have atropine on hand during an enucleation. Why is this?	During an enucleation procedure, the oculocardiac reflex may be stimulated during dissection of the ocular muscles resulting in bradycardia and/or asystole requiring the administration of atropine intravenously. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 176.*
What anesthetic technique is recommended for an enucleation.	General endotracheal anesthesia is recommended. The high incidence of postoperative nausea and vomiting also necessitates the use of antiemetics. *Jaffe RA, Samuels SI. Anesthesiologist's Manual of Surgical Procedures. 4th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2009: 162-164.*
Is deep extubation contraindicated on emergence from an enucleation?	No, because bucking and coughing on emergence can result in venous congestion and postoperative bleeding, deep extubation or administration of fentanyl and/or lidocaine should be considered. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 174-175.*
During opthalmic surgery, the surgeon requests that you administer mannitol. Why would he request this and how would you administer it?	The administration of mannitol may be used to decrease intraocular pressure intraoperatively. The total dose should not exceed 1.5-2 grams/kg over a 30-60 minute period. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 175.*
A patient presenting for opthalmic surgery is taking acetazolamide. Why would this patient be taking this drug and what electrolyte abnormalities may be present?	Acetazolamide is a carbonic anhydrase inhibitor that may be used to treat increased intraocular pressure. Because it also inhibits carbonic anhydrase in the kidneys, patients taking this on a chronic basis may exhibit acidosis, hypokalemia, and hyponatremia. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 175.*
What other conditions should you evaluate for preoperatively in a patient presenting for strabismus surgery?	Strabismus surgery is the most common pediatric ocular surgery and may be an indicator that there is underlying disease as it is associated with a high incidence of prematurity, cerebral palsy, and craniofacial abnormalities. Strabismus surgery is also associated with a higher than normal incidence of malignant hyperthermia. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1199.*

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Prodigy Anesthesia Quick Review

Extracranial Surgery

Why should hypercapnia be avoided in a patient undergoing strabismus surgery?	Hypercapnia increases the risk of bradycardia due to the oculocardiac reflex which can occur due to traction on the ocular muscles. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1199.*
What is the incidence of postoperative nausea and vomiting in patients undergoing strabismus surgery?	The incidence of postoperative nausea and vomiting is as high as 85% in this patient class. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1199.*
What precautions should you take during the anesthetic for a patient presenting for repair of an open-globe eye injury?	Deep extubation is recommended in this class of patient to decrease the risk of coughing and injuring the newly repaired eye. Succinylcholine can increase intraocular pressure and should be avoided in this patient. Retrobulbar and peribulbar blocks can increase the pressure exerted against the eye and increase the risk for extrusion of ocular contents in an open globe injury. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 768.*
Blockade of which nerve would prevent a patient from squinting during placement of a lid speculum for opthalmologic procedures?	Blockade of the facial nerve will prevent the patient from being able to squint the eyelids and allow placement of a lid speculum. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 767.*
Why should nitrous oxide be avoided in patients undergoing repair of a detached retina?	Opthalmologists may inject expanding gases such as sulfur hexafluoride or perfluoropropane into the eye to press a torn retina back into place. The use of nitrous oxide can cause the injected bubble to expand rapidly, causing a dramatic increase in intraocular pressure and possibly interrupting retinal blood flow. Nitrous oxide is often avoided in these surgeries, but if used, it should be discontinued at least 15 minutes before an expanding gas is injected. If a patient is presenting for a second surgery, nitrous oxide should be avoided for up to 10 days after an air injection, 30 days after the injection of sulfur hexafluoride and 90 days after the injection of C3F8. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 175.*

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Extracranial Surgery

During general anesthesia for endoscopic surgery of the ethmoid sinus, the surgeon indicates he may have pierced the ethmoid roof. What are the anesthesia implications you should consider on emergence?

The roof of the ethmoid sinus is the cribriform plate and anterior skull base. Piercing this sinus can result in a communication between the airway and the cranial vault and cerebrospinal fluid may be seen leaking from the nose. Bag-mask ventilation is to be avoided in any patients with a known skull base defect as it could result in pneumocephalus.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 230.

Why should nitrous oxide be avoided in patients undergoing repair of a a tympanoplasty?

Because nitrous oxide is more soluble than nitrogen, it can diffuse into an air-filled cavity faster than nitrogen can diffuse out or be absorbed into the blood. As a result, once the graft is placed, the inner ear becomes a closed space. The accumulation of pressure in the middle ear due to the diffusion of nitrous oxide into it can displace the graft. Because of the high incidence of postoperative nausea and vomiting associated with these cases, drugs used to prevent nausea such as 5-HT3 blockers, decadron, and even droperidol are commonly employed.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 784.

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